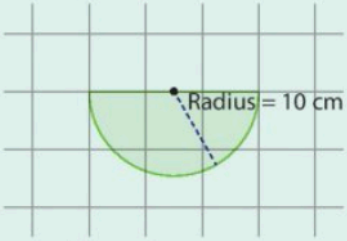
Example 3

Find the length of the perimeter of a semi-circular disc of radius 10 cm. ($\pi = 3.14$)

Give your answer correct to 2 significant figures.



Circumference of circle $= 2\pi r$

$c = 2 \times 3.14 \times 10 = 62.8$ cm

Circumference of ***semi***-circle = 31.4 cm
Length of diameter $= 2 \times 10 = 20$ cm
Length of perimeter $= 31.4 + 20 = 51.4$ cm = 51 cm correct to 2 significant figures.

Exercise 5C

1. Work out the length of the circumference of the circles with the following diameters.
Take $\pi = 3.14$ and give your answers correct to one place of decimals:

(i) 2 cm	(ii) 5.1 m	(iii) 2.5 mm	(iv) 15 m
(v) 4.5 cm	(vi) 25.1 mm	(vii) 0.5 km	(viii) 1.5 km

2. Calculate the length of the circumferences of the circles with the following radii:
Take $\pi = 3.14$ and give your answers correct to 3 significant figures.

(i) 5.1 cm	(ii) 25 cm	(iii) 2.8 mm	(iv) 15 m
(v) 14.5 m	(vi) 12.1 mm	(vii) 1.5 km	(viii) 2 km

Calculate the length of the circumference of each of the following, giving each answer correct to 3 significant figures: Use the π key on your calculator for questions 3–9

3. a circle with radius 19 cm.

4. a circle with diameter 24 m.

5. a circular table with a diameter 90 cm.

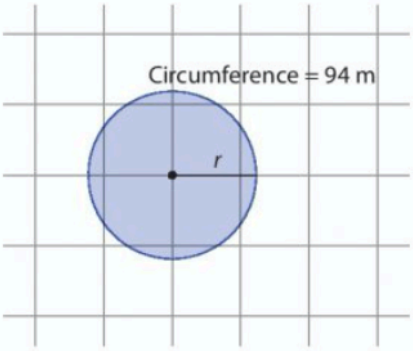
6. a circular flower bed of radius 1.5 m.

7. a circular flower pot of diameter 30 cm.

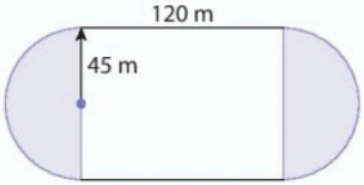
8. a pen of radius 0.5 cm.

9. a circular golf green of diameter 22 m.

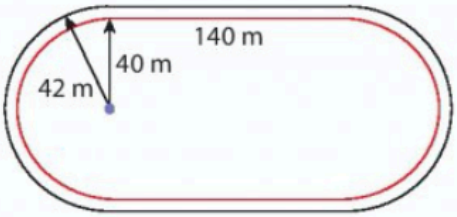
10. A circular pond has a circumference of 94 m.
Calculate the length of the radius correct to 2 significant figures. Take $\pi = 3.14$.



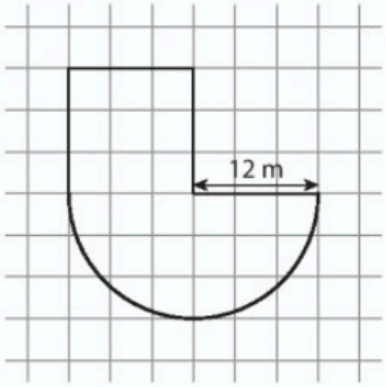
11. A running track has two semi-circular ends each of radius 45 m and two connecting straight sections of length 120 m.
Work out the length of the track, correct to the nearest m. Take $\pi = 3.14$.

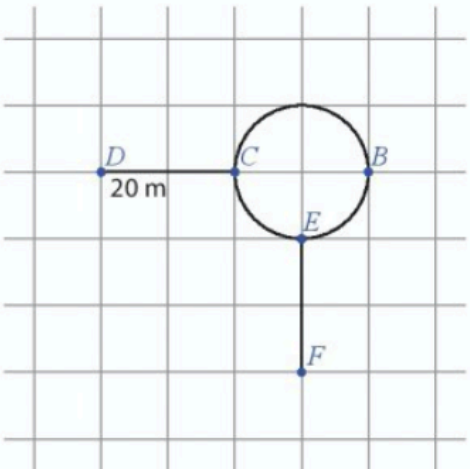


12. Peter runs on the inner lane (red) with semi-circular ends of radius 40 m.
Paul runs on the outer lane (black) with semi-circular ends of radius 42 m.
How much further has Paul to run?
Taking $\pi = 3.14$, correct your answer to the nearest metre.



13. Study this graphic comprising a square and a semi-circle and then using the measurement given, calculate the length of the perimeter of the shape, using the π key on your calculator. Give your answer correct to 1 decimal place.



14. Joanne bought 1 m of fine wire to make bead bracelets for her younger sister. Each bracelet is to have a diameter of 6 cm and she needs 1 cm for tying the ends together. How many bracelets can she make with the wire?

15. Two roads DC and EF are connected by a roundabout. If each square on a map of the roads has a side of 20 m, find the distance travelled going from D to F.